

SLEETMUTE, AK, USA

WMO: 703407

Lat: 61.717N

Lon: 157.150W

Elev: 178

StdP: 14.60

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26553

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.5	-30.9	-29.2	1.1	-21.3	-28.6	1.1	-20.3	15.4	25.5	12.7	15.6	0.3	10	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	74.7	58.6	71.8	57.3	68.0	56.0	61.5	70.9	59.3	67.3	57.5	65.2	3.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	70.4	64.0	55.2	65.3	62.0	54.1	62.6	60.8	27.3	70.2	25.9	67.4	24.7	65.4	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.5	10.0	8.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-28.5	63.6	3.4	2.9	-30.9	65.6	-32.9	67.3	-34.8	68.9	-37.2	71.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.9	-1.3	9.8	15.7	32.6	45.5	54.7	57.0	54.0	45.9	29.3	13.1	0.9		
	DBStd	23.55	18.52	15.43	12.95	8.67	7.01	5.08	4.81	4.59	5.59	10.40	12.81	15.69		
	HDD50	7919	1592	1126	1063	522	172	13	3	16	140	641	1108	1523		
	HDD65	12836	2057	1546	1528	972	603	312	253	341	572	1106	1558	1988		
	CDD50	565	0	0	0	0	34	153	220	139	18	1	0	0		
	CDD65	7	0	0	0	0	0	2	5	0	0	0	0	0		
	CDH74	158	0	0	0	0	7	56	93	2	0	0	0	0		
	CDH80	32	0	0	0	0	0	11	21	0	0	0	0	0		
(14) Wind	WSAvg	2.5	1.9	2.3	3.1	3.2	3.5	2.6	2.5	2.2	2.1	2.5	2.1	1.9		
(15) Precipitation	PrecAvg	27.3	2.0	1.7	1.5	1.5	1.6	2.2	3.1	3.5	3.1	2.5	2.3	2.3		
	PrecMax	33.3	4.4	3.6	3.3	2.4	3.0	3.1	4.6	5.3	5.4	4.8	5.1	3.7		
	PrecMin	21.3	0.6	0.2	0.3	0.4	0.4	1.1	1.5	1.5	0.9	0.5	0.2	0.8		
	PrecStd	3.1	0.9	0.7	0.7	0.5	0.6	0.6	0.7	0.9	1.2	1.0	1.2	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	42.1	44.9	57.4	74.3	80.6	82.5	72.9	66.0	53.1	42.9	37.2		
		MCWB	34.6	35.3	36.2	43.1	54.4	61.0	64.5	61.3	54.2	47.5	35.8	34.8		
	2%	DB	34.4	37.2	39.9	54.2	70.2	74.1	75.3	68.5	60.8	47.9	36.6	33.5		
		MCWB	32.2	32.7	33.6	41.6	52.6	57.3	60.7	58.1	50.8	44.4	33.4	31.6		
	5%	DB	31.5	34.4	36.9	50.4	64.4	71.4	72.0	64.8	57.1	45.5	34.0	29.3		
		MCWB	30.3	31.6	32.4	39.9	49.9	55.8	59.3	56.3	50.2	42.1	31.6	28.5		
	10%	DB	27.0	30.4	34.3	46.3	59.4	66.5	67.8	62.8	54.8	43.2	30.3	24.5		
		MCWB	24.9	28.5	31.1	37.9	46.5	54.3	57.7	55.1	49.2	40.0	29.2	22.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	35.5	38.0	44.4	56.1	63.0	66.3	63.2	55.4	48.3	36.6	35.3		
		MCDB	38.8	40.5	41.6	54.7	70.8	76.6	79.2	68.7	61.6	52.2	41.3	37.7		
	2%	WB	32.5	33.4	35.0	42.1	53.3	59.3	62.5	59.9	53.1	45.7	33.6	31.7		
		MCDB	34.6	36.8	38.4	52.1	67.5	70.3	73.1	65.5	57.5	47.6	36.6	33.2		
	5%	WB	30.3	31.9	33.0	40.3	50.6	57.2	60.2	58.0	51.4	42.9	31.8	28.4		
		MCDB	31.8	34.6	36.7	49.2	62.8	67.8	69.0	63.5	56.0	45.1	33.6	29.0		
	10%	WB	25.9	28.7	31.1	38.5	48.0	55.3	58.4	56.2	50.0	40.5	29.3	23.3		
		MCDB	27.8	30.8	34.4	46.2	58.2	65.7	66.2	61.2	54.5	43.2	30.3	25.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	17.6	18.8	18.5	19.7	19.0	15.8	14.4	14.3	11.3	12.2	14.4		
		MCDBR	14.9	15.7	17.6	25.2	28.1	26.8	25.7	20.1	18.4	12.7	11.2	15.7		
		MCWBR	13.1	13.3	13.2	15.3	14.7	12.9	13.0	11.9	11.7	9.2	9.6	14.7		
	5% WB	MCDBR	14.9	15.4	14.8	22.9	25.7	23.1	21.9	16.8	15.6	10.8	10.9	15.6		
		MCWBR	13.3	13.4	11.4	14.3	13.4	11.8	11.5	10.6	10.1	8.3	9.5	14.7		
(40) Clear-Sky Solar Irradiance	taub		0.220	0.268	0.275	0.343	0.366	0.376	0.376	0.381	0.318	0.292	0.228	N/A		
	taud		2.072	2.229	2.273	2.165	2.331	2.351	2.392	2.381	2.543	2.430	2.081	N/A		
	Ebn at Noon		156	225	270	266	271	269	266	253	255	218	147	N/A		
	Edh at Noon		11	20	27	38	35	35	33	31	22	17	11	N/A		
(44) All-Sky Solar Radiation	RadAvg		98	312	897	1433	1623	1666	1395	1088	731	333	122	47		
	RadStd		15	44	112	131	180	126	129	122	105	40	20	6		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+1.80	N/A	+1.53	N/A	N/A	N/A	-538	-644	N/A	N/A

Nomenclature: See separate page